

A THEORETICAL ANALYSIS

We analyze the main objectives in Secs. 4.2–4.3 from three aspects: the statistical meaning of the authenticity signal, the ordering induced by the hierarchical adversarial reward, and the geometric effect of the cluster-driven objective. The analysis characterizes the objectives optimized by T-Rex rather than claiming global convergence of the practical RL procedure.

A.1 Authenticity-Aware Exploration

The authenticity discriminator $S_\psi : \mathcal{Z} \rightarrow [0, 1]$ is trained to distinguish latent embeddings of observed data from constructed negative samples. Let $p_+(z)$ and $p_-(z)$ denote the corresponding latent distributions with class priors π_+ and π_- , where $\pi_+, \pi_- > 0$. We consider latent points satisfying $\pi_+ p_+(z) + \pi_- p_-(z) > 0$.

LEMMA 1 (AUTHENTICITY LIKELIHOOD RATIO). *At the population optimum, $S_\psi^*(z) = \frac{\pi_+ p_+(z)}{\pi_+ p_+(z) + \pi_- p_-(z)}$ and $\text{logit } S_\psi^*(z) = \log \frac{p_+(z)}{p_-(z)} + \log \frac{\pi_+}{\pi_-}$. Hence $R_{\text{auth}}(z) = \gamma_{\text{auth}} S_\psi^*(z)$ preserves the likelihood-ratio ordering for $\gamma_{\text{auth}} > 0$.*

PROOF. The population binary cross-entropy objective is

$$\mathcal{L}_{\text{auth}}(S) = -\pi_+ \int p_+(z) \log S(z) dz - \pi_- \int p_-(z) \log(1-S(z)) dz.$$

Since the objective is pointwise separable, for each fixed z it suffices to minimize

$$-\pi_+ p_+(z) \log S - \pi_- p_-(z) \log(1-S).$$

First consider the common support where $p_+(z) > 0$ and $p_-(z) > 0$. Differentiating with respect to S gives

$$-\frac{\pi_+ p_+(z)}{S} + \frac{\pi_- p_-(z)}{1-S}.$$

Setting the derivative to zero yields

$$S_\psi^*(z) = \frac{\pi_+ p_+(z)}{\pi_+ p_+(z) + \pi_- p_-(z)}.$$

The pointwise objective is strictly convex for $S \in (0, 1)$, since

$$\frac{\pi_+ p_+(z)}{S^2} + \frac{\pi_- p_-(z)}{(1-S)^2} > 0,$$

so the stationary point is the unique minimizer on the common support. If $p_-(z) = 0$ and $p_+(z) > 0$, the minimum is attained at $S_\psi^*(z) = 1$; if $p_+(z) = 0$ and $p_-(z) > 0$, it is attained at $S_\psi^*(z) = 0$. These boundary cases agree with the same expression for $S_\psi^*(z)$.

On the common support, rearranging the optimal discriminator gives

$$\frac{S_\psi^*(z)}{1-S_\psi^*(z)} = \frac{\pi_+ p_+(z)}{\pi_- p_-(z)}.$$

Taking logarithms yields

$$\text{logit } S_\psi^*(z) = \log \frac{p_+(z)}{p_-(z)} + \log \frac{\pi_+}{\pi_-}.$$

Since the sigmoid function is strictly increasing and $\gamma_{\text{auth}} > 0$, both $S_\psi^*(z)$ and $R_{\text{auth}}(z) = \gamma_{\text{auth}} S_\psi^*(z)$ preserve the ordering induced by $p_+(z)/p_-(z)$ on the common support. $\square$

Lemma 1 gives R_{auth} a direct statistical interpretation. At the population optimum, it ranks latent candidates according to how strongly they are supported by the observed-data latent distribution relative to the constructed negative distribution. It should therefore be interpreted as a relative plausibility signal rather than an exact test of membership in the unknown data manifold.

Moreover, writing $\sigma(\cdot)$ for the sigmoid function, the Phase-1 authenticity term can be expressed on the common support as

$$R_{\text{auth}}(z) = \gamma_{\text{auth}} \sigma\left(\log \frac{p_+(z)}{p_-(z)} + \log \frac{\pi_+}{\pi_-}\right)$$

at the population optimum. This makes explicit how the authenticity signal enters the exploration objective.

A.2 Confidence-Guided Risk Exploration

For a successful failure $\hat{y} \neq y^*$, Sec. 4.2 defines $R_{\text{adv}}(x, y^*) = \alpha_{\text{bonus}}(p)p$, where $p = P_f(\hat{y} | x)$ and $\alpha_{\text{bonus}}(p)$ follows Eq. (5). For $\hat{y} = y^*$, $R_{\text{adv}}(x, y^*) = -\beta_{\text{fail}} P_f(y^* | x)$, where $\beta_{\text{fail}} > 0$.

PROPOSITION 1 (HIERARCHICAL ORDERING OF R_{adv}). *For $\hat{y} \neq y^*$, R_{adv} is strictly increasing in $p = P_f(\hat{y} | x)$, and the three confidence intervals in Eq. (5) induce non-overlapping reward ranges. For $\hat{y} = y^*$, R_{adv} strictly decreases with $P_f(y^* | x)$.*

PROOF. Let $g(p) = \alpha_{\text{bonus}}(p)p$. According to Eq. (5),

$$g(p) = \begin{cases} 0.5p, & p \leq 0.5, \\ p, & 0.5 < p \leq 0.7, \\ 2p, & p > 0.7. \end{cases}$$

Within each confidence interval, g is linear with a strictly positive slope. Over the three nominal confidence intervals, the corresponding reward ranges are

$$g(p) \in \begin{cases} [0, 0.25], & p \leq 0.5, \\ (0.5, 0.7], & 0.5 < p \leq 0.7, \\ (1.4, 2], & p > 0.7. \end{cases}$$

These ranges are strictly ordered and non-overlapping. In particular,

$$g(0.5) = 0.25 < \inf_{p > 0.5} g(p) = 0.5,$$

and

$$g(0.7) = 0.7 < \inf_{p > 0.7} g(p) = 1.4.$$

Therefore, $p_1 > p_2$ implies $g(p_1) > g(p_2)$ both within the same confidence interval and across different intervals.

For an unsuccessful candidate, let $q = P_f(y^* | x)$. Since

$$R_{\text{adv}} = -\beta_{\text{fail}} q$$

with $\beta_{\text{fail}} > 0$, for any $q_1 > q_2$ we have

$$R_{\text{adv}}(q_1) < R_{\text{adv}}(q_2).$$

Hence the unsuccessful branch strictly decreases with the target model's confidence in y^* . $\square$

Proposition 1 shows that the hierarchical reward provides more than a binary success signal. Successful failures are strictly ordered by the confidence of the incorrect prediction, and the confidence tiers cannot reverse this ordering. Conversely, unsuccessful candidates receive larger penalties when the target model remains more confident in y^* .

Together with Lemma 1, the Phase-1 objective $R_{\text{base}} = R_{\text{auth}} + R_{\text{adv}}$ can therefore be interpreted as combining relative data plausibility with confidence-sensitive failure exploration.

A.3 Cluster-Driven Region Formation

Phase 1 explores high-reward failures without explicitly constraining their spatial organization. Phase 2 introduces R_{cluster} in Eq. (6) to impose region-level structure.

For the online clusters of $\mathcal{B}_t$, let

$$C_k = \{z_i \in \mathcal{B}_t : c(i) = k\}, \quad n_k = |C_k|,$$

where $n_k > 0$, and let the corresponding K-Means centroid be

$$\mu_k = \frac{1}{n_k} \sum_{z_i \in C_k} z_i.$$

Define

$$W(\mathcal{B}_t) = \frac{1}{B} \sum_{z_i \in \mathcal{B}_t} \|z_i - \mu_{c(i)}\|_2^2$$

and

$$D(\mathcal{B}_t) = \min_{j \neq k} \|\mu_j - \mu_k\|_2^2.$$

Then, for $\lambda_{\text{sep}} > 0$,

$$R_{\text{cluster}}(\mathcal{B}_t) = -W(\mathcal{B}_t) + \lambda_{\text{sep}} D(\mathcal{B}_t).$$

When referring below to a global maximizer of R_{cluster} , we assume that the maximum is attained over the feasible set of batches.

PROPOSITION 2 (GEOMETRY OF R_{cluster}). *For each C_k , μ_k minimizes its squared within-cluster distortion, and $W(\mathcal{B}_t) = \frac{1}{B} \sum_{k=1}^K \frac{1}{2n_k} \sum_{z_i, z_j \in C_k} \|z_i - z_j\|_2^2$. Moreover, if a feasible $\tilde{\mathcal{B}}_t$ satisfies $W(\tilde{\mathcal{B}}_t) \leq \eta$, $D(\tilde{\mathcal{B}}_t) \geq m^2$, and $\lambda_{\text{sep}} m^2 > \eta$, then any global maximizer $\mathcal{B}_t^*$ satisfies $D(\mathcal{B}_t^*) \geq m^2 - \eta/\lambda_{\text{sep}} > 0$.*

PROOF. For any u and cluster C_k ,

$$\begin{aligned} \sum_{z_i \in C_k} \|z_i - u\|_2^2 &= \sum_{z_i \in C_k} \|(z_i - \mu_k) + (\mu_k - u)\|_2^2 \\ &= \sum_{z_i \in C_k} \|z_i - \mu_k\|_2^2 + n_k \|\mu_k - u\|_2^2 \\ &\quad + 2(\mu_k - u)^\top \sum_{z_i \in C_k} (z_i - \mu_k). \end{aligned}$$

By the definition of the centroid,

$$\sum_{z_i \in C_k} (z_i - \mu_k) = 0.$$

Therefore,

$$\sum_{z_i \in C_k} \|z_i - u\|_2^2 = \sum_{z_i \in C_k} \|z_i - \mu_k\|_2^2 + n_k \|u - \mu_k\|_2^2.$$

Since the second term is nonnegative and vanishes if and only if $u = \mu_k$, the centroid μ_k minimizes the squared within-cluster distortion.

Next, consider the pairwise squared distances within C_k . The following sum is over ordered pairs $(z_i, z_j) \in C_k \times C_k$:

$$\begin{aligned} \sum_{z_i, z_j \in C_k} \|z_i - z_j\|_2^2 &= \sum_{i,j} \|(z_i - \mu_k) - (z_j - \mu_k)\|_2^2 \\ &= \sum_{i,j} \|z_i - \mu_k\|_2^2 + \sum_{i,j} \|z_j - \mu_k\|_2^2 \\ &\quad - 2 \sum_{i,j} (z_i - \mu_k)^\top (z_j - \mu_k). \end{aligned}$$

The first two terms each equal

$$n_k \sum_{z_i \in C_k} \|z_i - \mu_k\|_2^2,$$

while the cross term satisfies

$$\sum_{i,j} (z_i - \mu_k)^\top (z_j - \mu_k) = \left\| \sum_{z_i \in C_k} (z_i - \mu_k) \right\|_2^2 = 0.$$

Hence

$$\sum_{z_i, z_j \in C_k} \|z_i - z_j\|_2^2 = 2n_k \sum_{z_i \in C_k} \|z_i - \mu_k\|_2^2.$$

Equivalently,

$$\sum_{z_i \in C_k} \|z_i - \mu_k\|_2^2 = \frac{1}{2n_k} \sum_{z_i, z_j \in C_k} \|z_i - z_j\|_2^2.$$

Summing over k and dividing by B yields

$$W(\mathcal{B}_t) = \frac{1}{B} \sum_{k=1}^K \frac{1}{2n_k} \sum_{z_i, z_j \in C_k} \|z_i - z_j\|_2^2.$$

Thus, under the current K-Means assignments, the cohesion term is exactly the normalized within-cluster pairwise squared dispersion.

Now let $\mathcal{B}_t^*$ be any global maximizer of R_{cluster} over the feasible batch set. Since $\tilde{\mathcal{B}}_t$ is feasible,

$$\begin{aligned} R_{\text{cluster}}(\mathcal{B}_t^*) &\geq R_{\text{cluster}}(\tilde{\mathcal{B}}_t) \\ &= -W(\tilde{\mathcal{B}}_t) + \lambda_{\text{sep}} D(\tilde{\mathcal{B}}_t) \\ &\geq -\eta + \lambda_{\text{sep}} m^2. \end{aligned}$$

On the other hand, since $W(\mathcal{B}_t^*) \geq 0$,

$$\begin{aligned} R_{\text{cluster}}(\mathcal{B}_t^*) &= -W(\mathcal{B}_t^*) + \lambda_{\text{sep}} D(\mathcal{B}_t^*) \\ &\leq \lambda_{\text{sep}} D(\mathcal{B}_t^*). \end{aligned}$$

Combining the two inequalities gives

$$\lambda_{\text{sep}} D(\mathcal{B}_t^*) \geq -\eta + \lambda_{\text{sep}} m^2.$$

Since $\lambda_{\text{sep}} > 0$,

$$D(\mathcal{B}_t^*) \geq m^2 - \frac{\eta}{\lambda_{\text{sep}}}.$$

Finally, the condition $\lambda_{\text{sep}} m^2 > \eta$ implies

$$m^2 - \frac{\eta}{\lambda_{\text{sep}}} > 0.$$

Hence the global optimum of the clustering component retains a positive minimum inter-centroid separation and cannot exhibit complete centroid collapse. $\square$

Proposition 2 gives complementary interpretations of the two terms in Eq. (6). The cohesion term is exactly equivalent to reducing within-cluster pairwise dispersion. For a global optimum of the clustering objective, the separation term further provides a quantitative lower bound on the minimum centroid distance whenever a sufficiently compact and separated feasible configuration exists.

Substituting the pairwise identity into the clustering contribution of Eq. (7) gives the exact reformulation

$$\lambda_{\text{cluster}} R_{\text{cluster}}(\mathcal{B}_t) = -\frac{\lambda_{\text{cluster}}}{B} \sum_{k=1}^K \frac{1}{2n_k} \sum_{z_i, z_j \in C_k} \|z_i - z_j\|_2^2 + \lambda_{\text{cluster}} \lambda_{\text{sep}} D(\mathcal{B}_t).$$

Thus Phase 2 augments the Phase-1 risk–authenticity objective with an explicit penalty on within-region pairwise dispersion and a reward on the minimum separation between generated regions. The proposition characterizes the geometry induced by the clustering component in latent space; it does not by itself imply topological connectedness, semantic equivalence in the original tabular feature space, or global convergence of the full Phase-2 RL objective.